My “Secure” Home Network

How a quest for more speed turned into VLANs, multiple WiFi networks, local automation and a lot more security

By Michael Coar

I didn’t set out to build a highly segmented home network. I just wanted my AI models to download faster. Then, as these projects tend to do, things got a little out of hand.

My initial foray into rebuilding my home network wasn’t really about security. It was about speed.

I was getting deeper into AI and home-lab projects — a whole other story — and discovered that AI models are enormous. I wanted faster Internet downloads, but I also wanted much faster transfers inside the house, particularly between my local AI server and my workstation.

So I started upgrading. An Ubiquiti Dream Machine Pro became the heart of the network, along with managed switches and high-speed Ethernet. Then came UniFi wireless access points, MoCA, multiple VLANs, multiple WiFi networks and Technitium DNS.

In other words: I wanted faster downloads. I somehow ended up designing a small corporate network in my house.

The Physical Network: Wire It When You Can

Wherever practical, I use wired Ethernet. It is fast, predictable and avoids many of the variables that come with WiFi.

In a couple of places where running new Ethernet wasn’t practical, the house already had coaxial cable from the cable-TV era. Rather than let that wiring sit unused, I installed MoCA adapters and turned the old TV coax into network connections. MoCA allows you to run your network on coax cable, with speeds up to 2.5Gb.

I also use managed switches. Besides giving me high-speed wired connectivity, they let me decide which logical network — or VLAN — an individual Ethernet port belongs to. That means a wired streaming box doesn’t automatically have to live on the same network as my computers.

WiFi: Four Access Points, Several Different Jobs

For wireless coverage, I use four UniFi U7 Pro XG wireless access points: one in the living room, one in the bedroom, one in the dining room and one mounted under the eave in the backyard.

The goal isn’t simply strong WiFi everywhere. The access points also let me present different WiFi networks for different types of devices and map those networks to the appropriate VLAN.

Not every device that knows my WiFi password deserves to meet every other device in my house.

The Four Main Network Segments

Main network (.0)

This is the trusted side of the house: my computers, phone, iPads, network attached storage and other primary computing devices. These are the systems that contain my files, credentials and day-to-day work, so they don't need to share a neighborhood with every cloud-connected gadget I own.

Fire TV network (.1)

I still have a couple of Amazon Fire TV devices, although I intend to phase them out. One is wired and two are is wireless. They ended up on their own segment because the wireless devices needed 5 GHz WiFi rather than the 2.4 GHz IoT network I had originally intended to use.

This is a good example of what happens in the real world: the elegant network diagram meets an actual consumer device, and the consumer device wins.

IoT network (.2)

This is where the cloud-connected household devices live: my Arlo video doorbell, Ecobee thermostats, Abode security system and cameras, plus a few Raspberry Pi devices — including the Raspberry Pi behind the magic mirror in my front hall.

The IoT WiFi is 2.4 GHz, which is still common for smart-home devices and provides good range. More importantly, these devices are separated from my trusted computers.

Guest network (.3)

Guests get their own network. Their devices can reach the Internet without joining the same network as my computers, servers and other trusted equipment.

That isn't an indictment of my friends. I like my friends. I simply have no idea when they last updated their laptops.

The WiFi Networks Behind It

The result is more WiFi names than most sane households probably need, but each has a specific purpose:

- Main WiFi — 2.4, 5 and 6 GHz for my primary computing devices on the trusted .0 VLAN.
- 5 GHz WiFi — a temporary dedicated network for trusted devices that do not roam or band-steer well. I expect to retire it on the .0 VLAN.
- 6 GHz WiFi — another temporary dedicated network for main devices that behave better when explicitly placed on 6 GHz also on the .0 VLAN.
- IoT WiFi — 2.4 GHz for cloud-connected smart-home devices on the .2 IoT VLAN.
- IoT B WiFi — 5 GHz for the wireless Fire TV devices on the .1 VLAN.
- HA WiFi — 2.4 GHz for my Home Assistant environment and local home-automation devices on the .0 VLAN
- Guest WiFi — 2.4, 5 and 6 GHz, mapped to the isolated .3 guest VLAN.

Moving Home Automation Back Inside the House

The HA network is part of a larger project: I am gradually moving away from cloud-dependent home automation and toward Home Assistant and locally controlled devices.

The objective is simple. If a light switch, sensor or automation can operate entirely inside my house, I would rather not require an outside cloud service simply to turn something on or off.

Ultimately, I want as many home-automation devices as practical to communicate locally and have no reason to reach the Internet at all.

If the Internet goes down, I would still like the house to remember how a light switch works.

That project deserves its own article, so I'll save the details for another day.

Technitium: DNS, Ad Blocking and Visibility

Technitium DNS currently sits at the center of my DNS setup. The router points to Technitium as its DNS server, so devices using the router's normal DNS configuration ultimately send their DNS requests through Technitium. This running on a RaspberryPi 5 with 16Gb of RAM, and an NVMe hat that hold a 512GB NVMe drive.

That gives me network-level ad and tracker blocking without installing a browser extension on every device. More importantly, it gives me visibility into DNS activity — which is useful when a supposedly boring device suddenly becomes very chatty.

There are two changes I still plan to make.

Let the guests see the ads

The guest network currently uses Technitium, which means guests also inherit my blocking rules. That sounds wonderful until somebody wants to watch a video or visit a site that doesn't behave properly when an advertising domain is blocked.

My plan is to let the guest network use a conventional external DNS resolver instead. Guests get Internet access and network isolation; I get fewer requests to fix somebody else's CNN video.

Identify individual devices more precisely

Today, many client DNS requests are forwarded through the router, which can make it harder to identify the original device in Technitium's logs. My longer-term goal is to configure as many devices as practical to use Technitium directly.

That should make it easier to distinguish the workstation from the laptop from the AI server when analyzing DNS activity. Some IoT devices won't allow custom DNS settings, so this will never be universal, but greater visibility is the goal.

So, Is This Overkill?

For most households? Absolutely.

For mine? I don't think so.

I started this project because I wanted speed. What I ended up with is a network that is faster, easier to observe and far more segmented than the typical home setup.

My computers don't need to share a network with my Fire TVs. My security cameras don't need unrestricted access to my workstation. Guests don't need to be on the same network as my servers. And eventually, many of my home-automation devices shouldn't need the Internet at all.

There are still things I want to change, simplify and retire. In fact, a good network is probably never completely finished. Devices change. Standards change. Security concerns change. And occasionally you discover that one stubborn streaming box has very strong opinions about which WiFi band it wants to use.

But the guiding principle has become surprisingly simple:

Give every device the connectivity it needs — and no more.

Not bad for a project that started because I was impatient waiting for an AI model to download.